Completeness of Phase-free ZX

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Schedule

266. Exactly three mutually unbiased bases in dimension six. Proves $N(6) = 3$, resolving Zauner's dimension-six mutually unbiased bases conjecture: three such bases exist in $\mathbb{C}^6$, but four cannot. The exclusion is a complete certified computation under the stated binary64 arithmetic and compiler conditions. An independent companion proves the Matolcsi–Ruzsa–Weiner Fourier-vanishing conjecture for order-six complex Hadamard matrices outside Tao's cubic equivalence class. (Lean)

The maximum number of mutually unbiased bases in dimension six

We prove that the maximum number of mutually unbiased orthonormal bases in $\mathbb{C}^6$ is three, resolving Zauner's dimension-six MUB conjecture. The upper bound is computer-assisted: under the stated binary64 arithmetic and compiler conditions, a complete execution of the documented verification pipeline excludes four arbitrary complex bases.



Accepted!

Schedule

$$[\![\cdot]\!] : \mathsf{ZX} \longrightarrow \mathsf{LinRel}_{\mathbb{F}_2}$$

Last time...

- **Universality** (full)

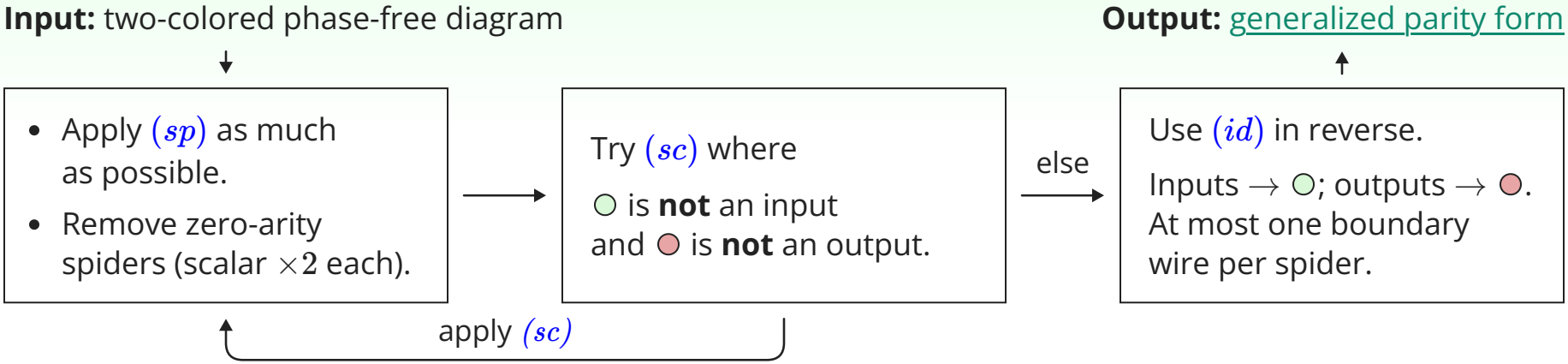
$$\forall M \in \mathsf{LinRel}_{\mathbb{F}_2}(m,n), \quad \exists D : m \to n \quad \text{s.t.} \quad [\![D]\!] = M.$$

This time...

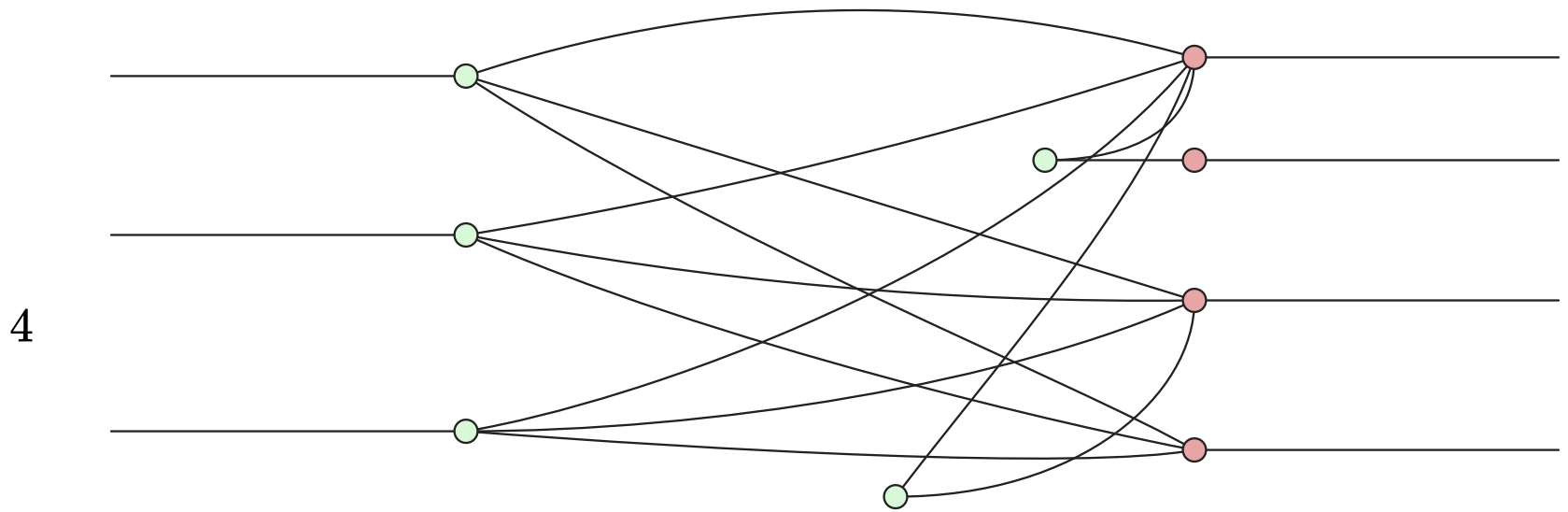
- **Completeness** (faithful)

$$\forall D_1, D_2 : m \to n, \quad [\![D_1]\!] = [\![D_2]\!] \quad \Longrightarrow \quad \mathcal{R} \vdash D_1 = D_2.$$

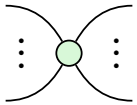
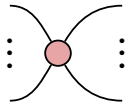
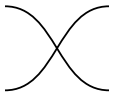
Algorithm 2: Reducing to Generalized Parity Form



Example.



Phase-free ZX

$Z_{n,m} : n \to m$ 			$X_{n,m} : n \to m$ 	
				
$\sigma : 2 \to 2$	$\mathrm{id}_1 : 1 \to 1$	$e : 0 \to 0$	$\eta : 0 \to 2$	$\epsilon : 2 \to 0$

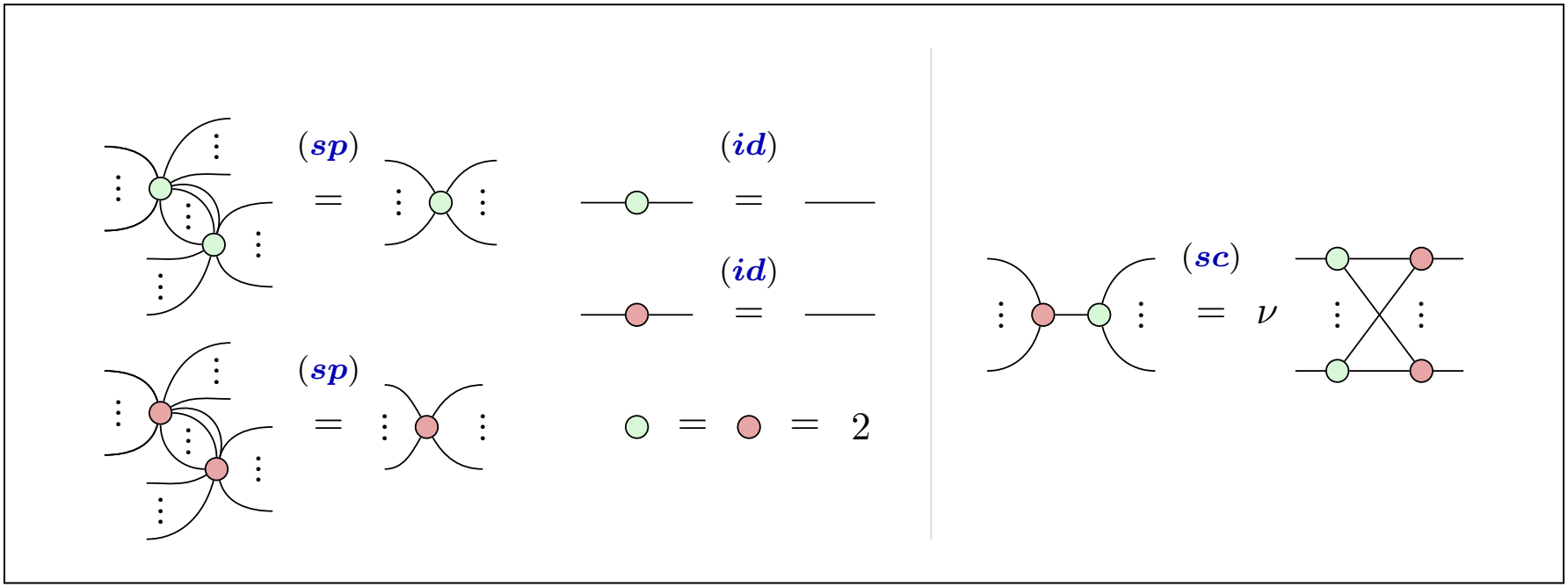
Generators of the Phase-free ZX calculus, $\mathcal{G}$.

Definition. The phase-free ZX calculus is the PROP

$$\mathsf{ZX} = \langle \mathcal{G} \mid \mathcal{R} \rangle.$$

Phase-free ZX

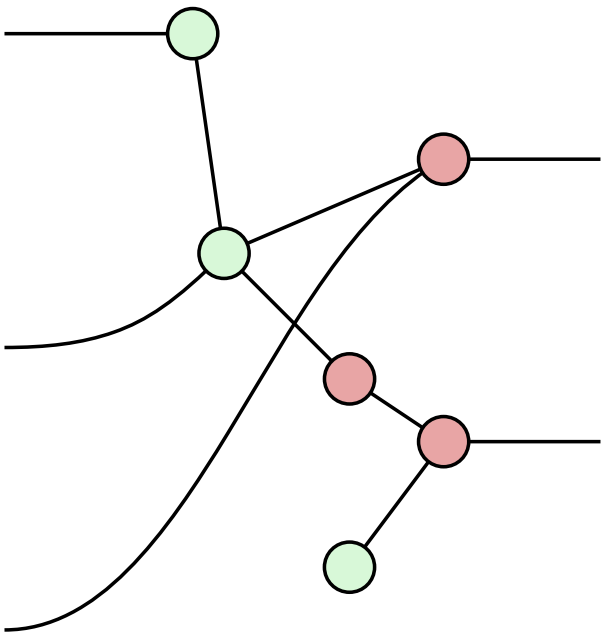
Figure 4.1.



Axioms of the phase-free zx calculus, $\mathcal{R}$.
where $\nu = 2^{(n-1)(m-1)/2}$.

+ only connectivity matters

Only connectivity matters



Standard Interpretation

$$[\![\cdot]\!] : \mathbf{ZX} \longrightarrow \mathbf{Qubit}$$

We define the standard interpretation inductively as follows:

$$[\![D_1 \circ D_2]\!] = [\![D_1]\!] \circ [\![D_2]\!], \qquad [\![D_1 \otimes D_2]\!] = [\![D_1]\!] \otimes [\![D_2]\!].$$

$$[\![\text{—}]\!] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad [\![\text{X}]\!] = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad [\![\square]\!] = 1$$

$$[\![\cup]\!] = |00\rangle + |11\rangle \qquad [\![\cap]\!] = \langle 00| + \langle 11|$$

$$[\![\vdots \bullet \vdots]\!] = |0\rangle^{\otimes m}\langle 0|^{\otimes n} + |1\rangle^{\otimes m}\langle 1|^{\otimes n}$$

$$[\![\vdots \circ \vdots]\!] = |+\rangle^{\otimes m}\langle +|^{\otimes n} + |-\rangle^{\otimes m}\langle -|^{\otimes n}$$

$$(\text{where } |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}})$$

Completeness

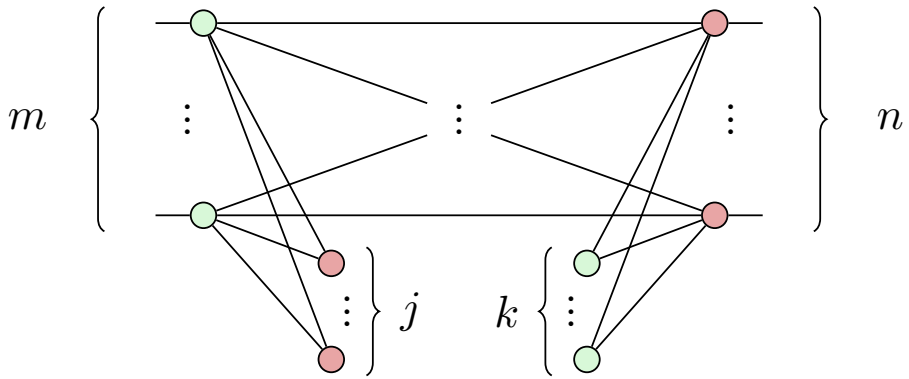
Theorem 4.3.6.

If two phase-free ZX-diagrams $D_1, D_2 : m \to n$ describe the same linear relation in $\mathbf{LinRel}_{\mathbb{F}_2}$, then $\mathcal{R} \vdash D_1 = D_2$.

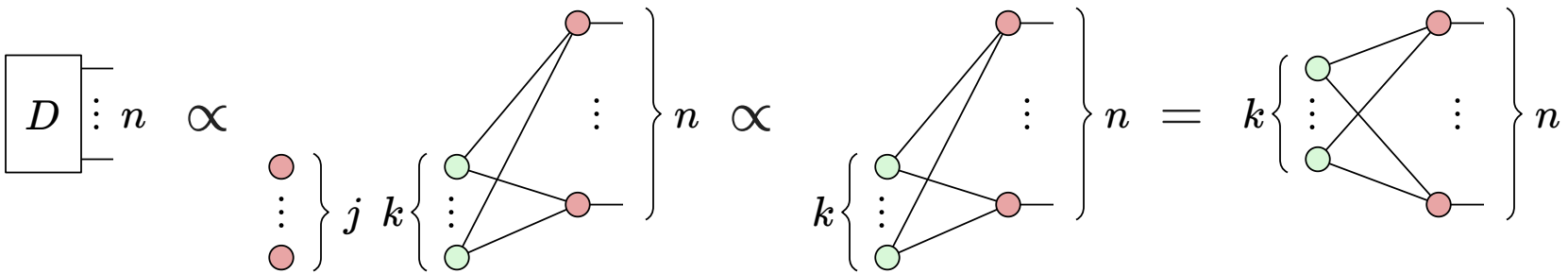
$$[\![D_1]\!] = [\![D_2]\!] \quad \Longrightarrow \quad \mathcal{R} \vdash D_1 = D_2.$$

States

Recall: Every phase-free ZX diagram can be rewritten to **generalised parity form**:

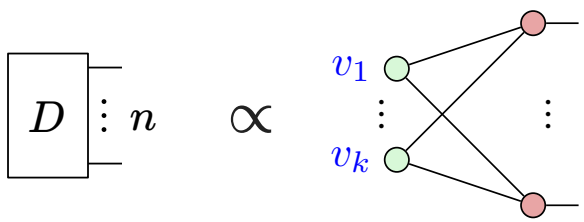


So, for a state $D : 0 \to n$,



State of the states...

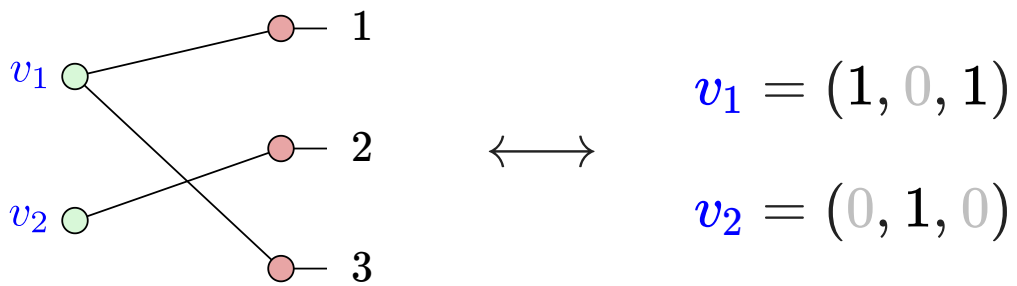
By the previous slide, any phase-free state can be written in the following way:



(This is a bipartite graph.)

So, we can describe a state by vectors $\{v_1, \ldots, v_k\}$ over $\mathbb{F}_2$, where $(v_p)_q = 1$ if and only if the p-th Z spider is connected to the q-th X spider.

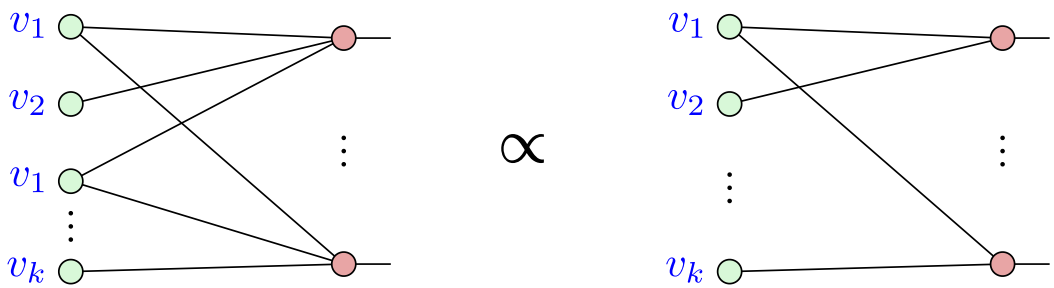
Example.



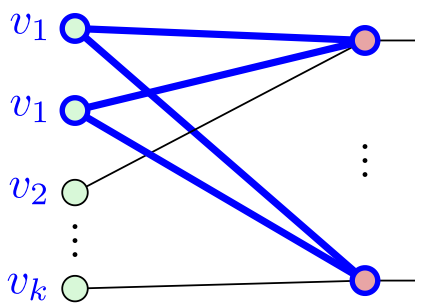
Duplicate vectors

Lemma *4.7.

We can always remove green spiders corresponding to duplicate bit-vectors.



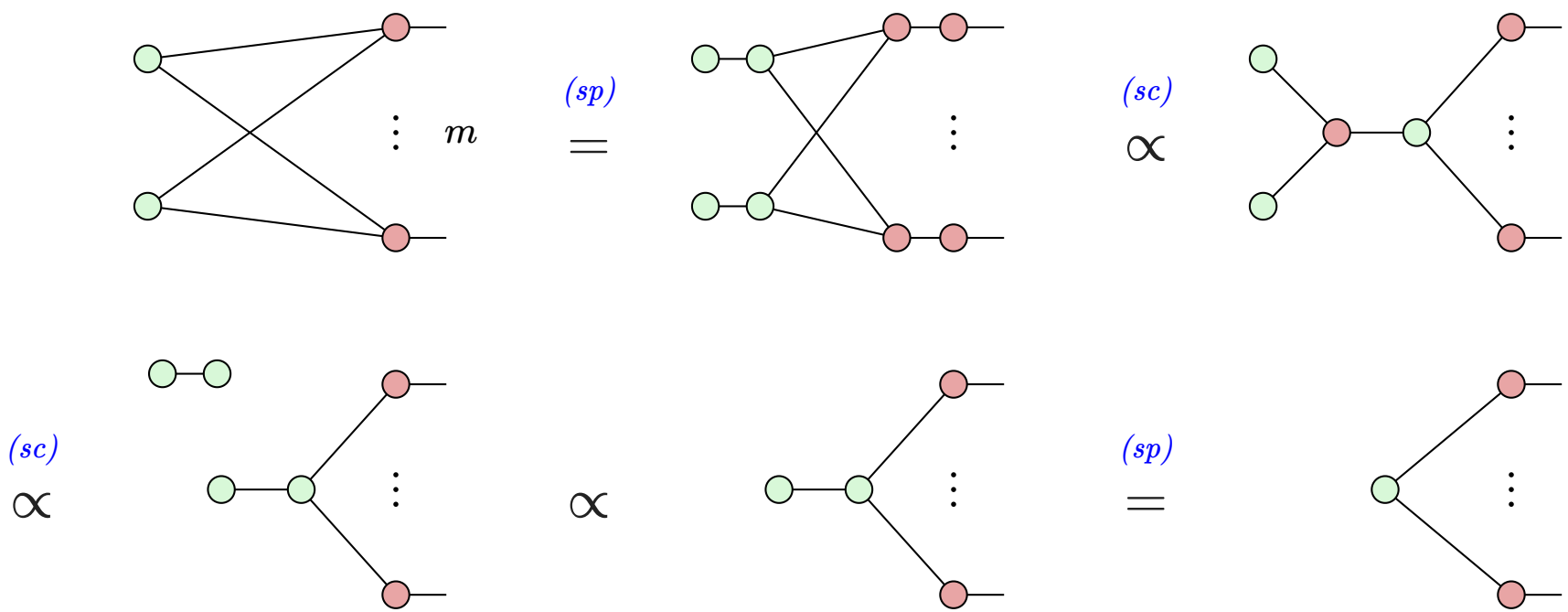
Proof. First, use connectivity to put the duplicate spiders together, say v_1, v_1.



Duplicate vectors

continued...

Now consider only the two v_1 spiders and their $m \leq n$ common red neighbours. This is a complete bipartite graph.



Linear subspaces

So, Z-X normal form certainly isn't unique!

We'll see that the important part is actually the span of our vectors $\{v_1, \ldots, v_k\}$, rather than the specific vectors themselves...

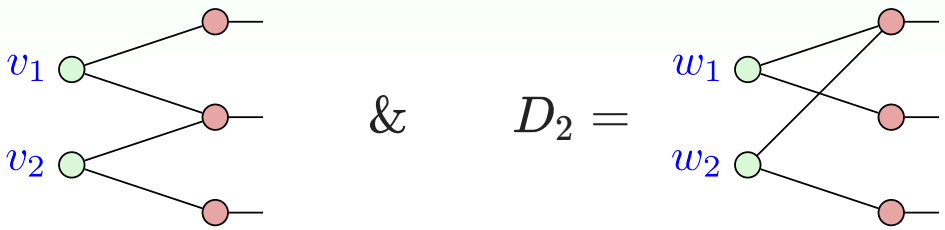
Recall:

$$\mathrm{Span}\{v_1, \ldots, v_k\} = \{a_1 v_1 + \cdots + a_k v_k \mid a_1, \ldots, a_k \in \mathbb{F}_2\}.$$

Nonuniqueness

Example.

$$D_1 = \quad \& \quad D_2 =$$



So,

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \; v_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \quad \& \quad w_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \; w_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

Note:

$$\mathrm{Span}\{v_1, v_2\} = \underbrace{\left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}}_{S} = \mathrm{Span}\{w_1, w_2\}$$

And,

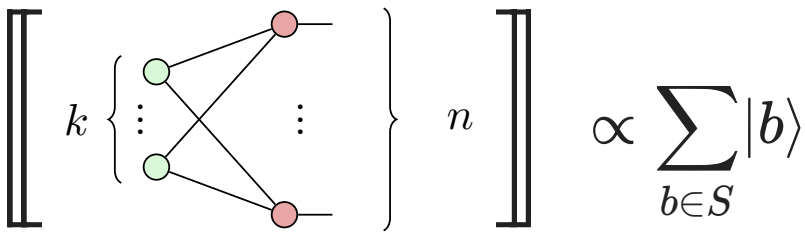
$$[\![D_1]\!] \propto \sum_{b \in S} |b\rangle \propto [\![D_2]\!]$$

Interesting!

Linear subspaces

Theorem 4.3.2.

For any Z-X normal form described by a set of $\mathbb{F}_2$-vectors $\{v_1, \ldots, v_k\}$, we have:



$$\propto \sum_{b \in S} |b\rangle$$

where $S = \mathrm{Span}\{v_1, \ldots, v_k\}$.

Proof of Theorem 4.3.2

$$\left[\!\!\left[\; k\left\{ \; \vdots \quad \vdots \; \right\} n \; \right]\!\!\right] \overset{(sp)}{=} \left[\!\!\left[\; k\left\{ \; \vdots \quad \vdots \quad \vdots \; \right\} n \; \right]\!\!\right]$$

$$= \left(\left[\!\!\left[\; \bullet\!\!- \; \right]\!\!\right] \right)^{\otimes k} \circ \left[\!\!\left[\; k\left\{ \; \vdots \quad \vdots \; \right\} n \; \right]\!\!\right]$$

$$= \left(|0\rangle + |1\rangle \right)^{\otimes k} \circ \left[\!\!\left[\; k\left\{ \; \vdots \quad \vdots \; \right\} n \; \right]\!\!\right]$$

Proof of Theorem 4.3.2

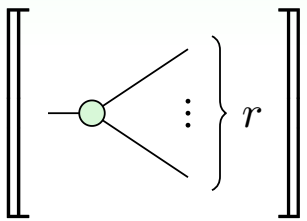
$$\left(|0\rangle + |1\rangle\right)^{\otimes k} \circ \left[\!\!\left[\; k\left\{ \; \right. \right. \right. \; \left. \left. \left. \right\} n \; \right]\!\!\right] = \left(\sum_{a \in \mathbb{F}_2^k} |a\rangle \right) \circ \left[\!\!\left[\; k\left\{ \; \right. \right. \right. \; \left. \left. \left. \right\} n \; \right]\!\!\right]$$

$$= \sum_{a \in \mathbb{F}_2^k} |a\rangle \circ \left[\!\!\left[\; k\left\{ \; \right. \right. \right. \; \left. \left. \left. \right\} n \; \right]\!\!\right]$$

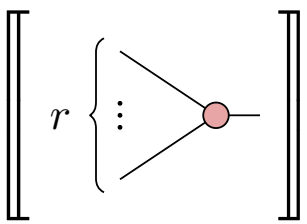
Proof of Theorem 4.3.2

Recall:

$$|x\rangle \circ \left[\!\!\left[\; \right\} r \; \right]\!\!\right] = |x\rangle^{\otimes r} \qquad \text{(COPY)}$$



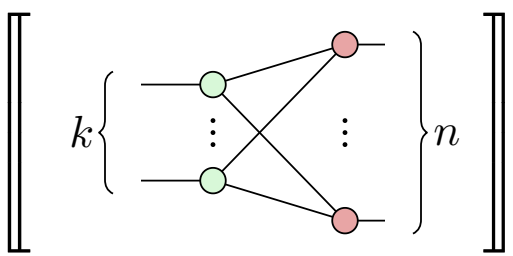
$$|x_1 \cdots x_r\rangle \circ \left[\!\!\left[\; \right\} r \; \right]\!\!\right] = 2^{(1-r)/2}|x_1 \oplus \cdots \oplus x_r\rangle \qquad \text{(XOR)}$$



So, for $f(a) = \sum_{i=1}^{k} a_i v_i$, with $\lambda \neq 0$,

For each red output j, $\displaystyle\bigoplus_{i:(v_i)_j=1} a_i = \sum_{i=1}^{k} a_i (v_i)_j = f(a)_j.$

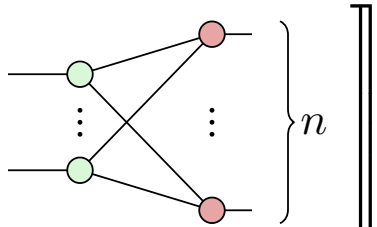
$$\implies |a\rangle \circ \left[\!\!\left[\; k \left\{ \; \right\} n \; \right]\!\!\right] = \lambda \bigotimes_{j=1}^{n} |f(a)_j\rangle = \lambda |f(a)\rangle \quad (\star)$$



Proof of Theorem 4.3.2

So,

$$\sum_{a\in\mathbb{F}_2^k}|a\rangle\circ\left[\!\!\left[\ k\left\{\ \vdots\quad\vdots\ \right\}n\ \right]\!\!\right]$$



$$\begin{aligned}
&\overset{(\star)}{=} && \lambda\sum_{a\in\mathbb{F}_2^k}|f(a)\rangle\\
&= && \lambda\sum_{b\in S}\left|f^{-1}(b)\right||b\rangle\\
&= && \lambda\,|\ker f|\sum_{b\in S}|b\rangle\\
&\propto && \sum_{b\in S}|b\rangle.
\end{aligned}$$

■

Changing a spanning set

Recall from linear algebra:

Proposition 4.3.3.

Let $\mathcal{B}$ and $\mathcal{C}$ be two sets spanning the same $\mathbb{F}_2$-linear subspace S. Then $\mathcal{B}$ can be transformed into $\mathcal{C}$ by

1. adding or removing the zero vector, or
2. replacing a pair of vectors $v, w \in \mathcal{B}$ with the pair $v, v + w$ zero or more times.

Proof.

Corollary 20.5.7. Row space versus row equivalence.

■

Changing the spanning set

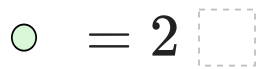
Lemma 4.3.4.

Let $\mathcal{B} = \{v_1, \ldots, v_k\}$ and $\mathcal{C} = \{w_1, \ldots, w_l\}$ describe two Z-X normal forms on n qubit space. If the sets $\mathcal{B}$ and $\mathcal{C}$ span the same subspace of $\mathbb{F}_2^n$, one can be transformed into the other, up to a scalar factor.

Proof.

We will show that the Z-X normal form for $\mathcal{B}$ can be transformed into that for $\mathcal{C}$ using the two cases described in Proposition 4.3.3.

First, note that adding or removing a zero vector corresponds to adding or removing a zero-legged Z spider, which simply adds or removes a scalar factor of 2.



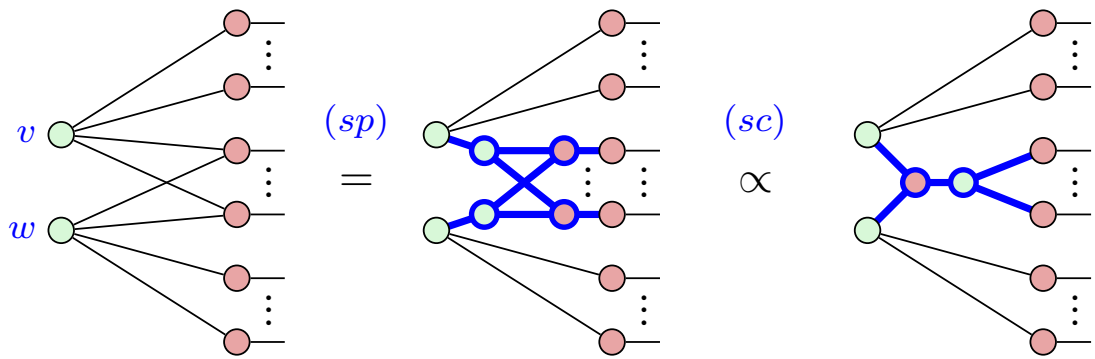
Adding the vectors

continued...

Hence, it suffices to show that we can transform a pair of Z-spiders described by the vectors v, w into Z spiders described by $v, v + w$. That is, we can "add" the wires of one spider to the other one, modulo 2.

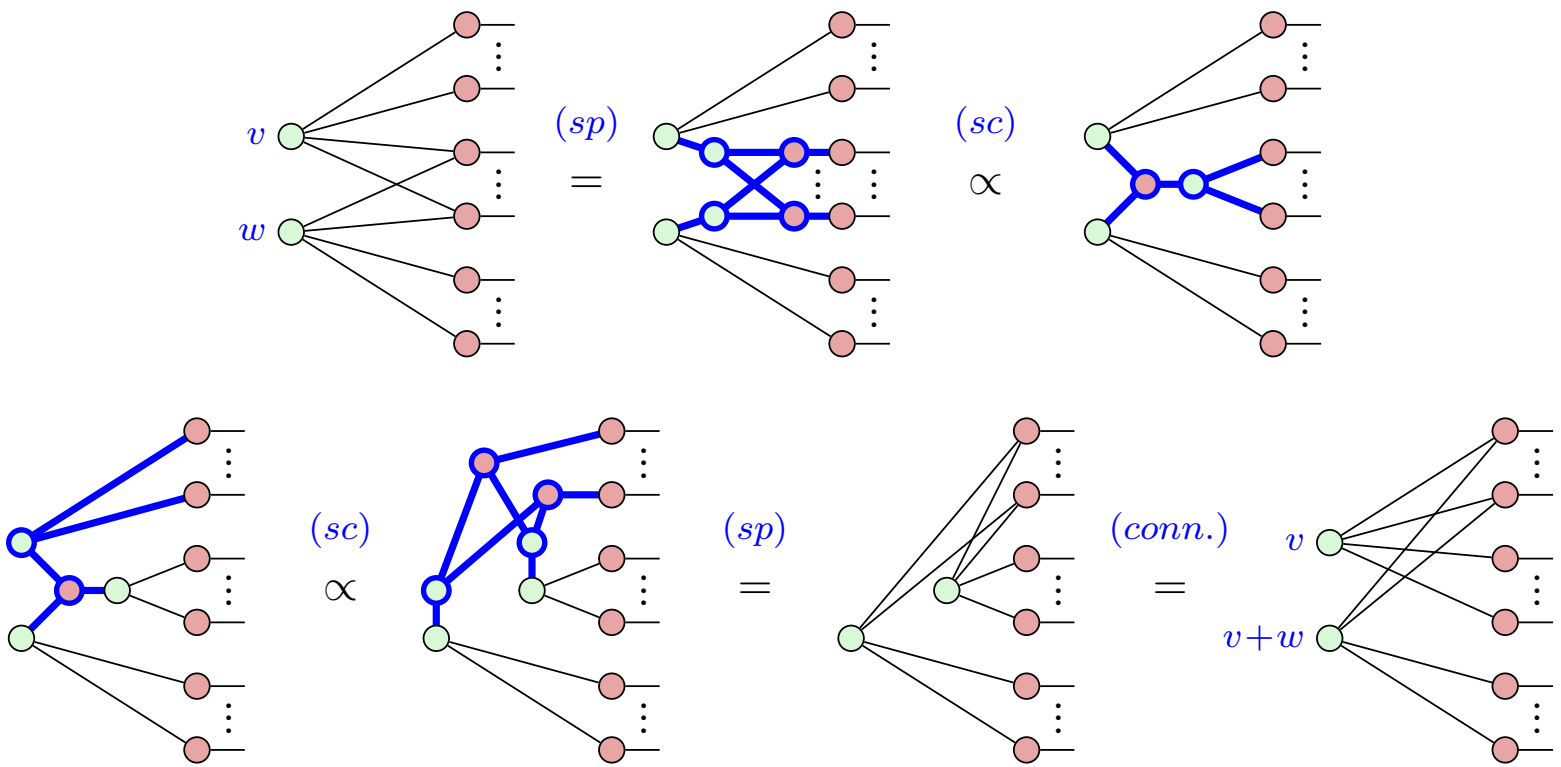
We can see this by splitting the X spiders into 3 groups: those connected to v, those connected to w, and those connected to both.

We can then perform the vector addition by two applications of the (sc) rule. First, we apply the rule in reverse:



Adding the vectors

continued...



Completeness for states

Proposition 4.3.5.

If two phase-free ZX-diagram states $D_1, D_2 : 0 \to n$ describe the same linear relation in $\mathbf{LinRel}_{\mathbb{F}_2}$, then $\mathcal{R} \vdash D_1 = D_2$.

$$[\![D_1]\!] = [\![D_2]\!] \quad \Longrightarrow \quad \mathcal{R} \vdash D_1 = D_2.$$

Proof.

Both diagrams can be brought into Z-X normal form.

By Theorem 4.3.2, their associated $\mathbb{F}_2$-vectors span the same subspace S.

Hence, by Lemma 4.3.4, one can be transformed into the other.

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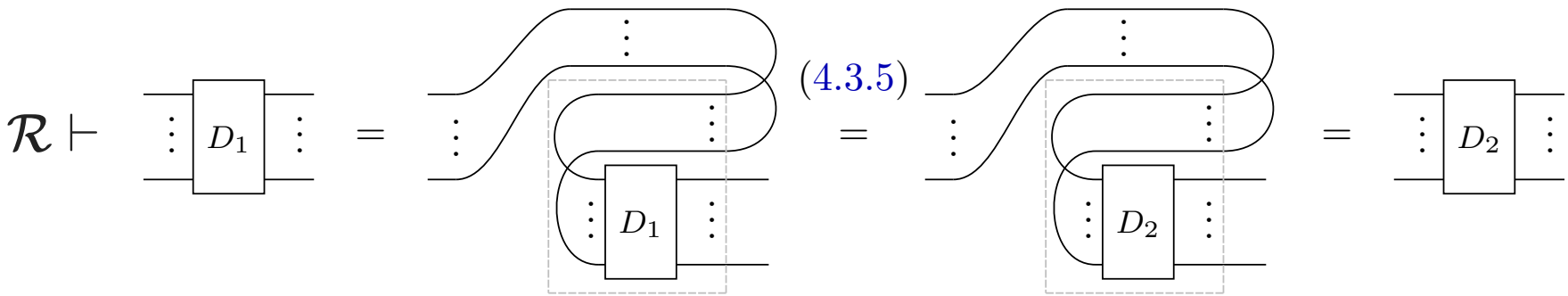
Completeness of phase-free ZX

Theorem 4.3.6.

If two phase-free ZX-diagrams $D_1, D_2 : m \to n$ describe the same linear relation in $\mathbf{LinRel}_{\mathbb{F}_2}$, then $\mathcal{R} \vdash D_1 = D_2$.

$$[\![D_1]\!] = [\![D_2]\!] \quad \Longrightarrow \quad \mathcal{R} \vdash D_1 = D_2.$$

Proof.



■

Thank you!